Supplemental Table S1: Search strategy for the electronic literature search

	Search strategy	Hits
PubMed (from inception to 02.06.2020)		
#1	(comorbidity OR comorbidities OR multimorbidity OR "chronic diseases"[Title/Abstract] OR "chronic conditions"[Title/Abstract] OR "chronic illnesses"[Title/Abstract] OR comorbidity[MeSH])	291,079
#2	((chemotherapy OR "adjuvant treatment" OR treatment[Title/Abstract] OR "adjuvant therapy" OR chemotherapy[MeSH]) AND (use OR uptake OR receipt OR nonuse OR underuse OR initiation OR administration OR utilization OR adherence OR patterns))	4,149,289
#3	("colon cancer" OR "bowel cancer" OR "colonic cancer" OR "colonic neoplasms"[MeSH])	100,026
	All combined: #1 AND #2 AND #3	*426*
Web of Science (from inception to 02.06.2020)		
#1	ALL=(comorbidity OR comorbidities OR multimorbidity OR "chronic diseases" OR "chronic conditions" OR "chronic illnesses")	165,825
#2	ALL=((chemotherapy OR "adjuvant treatment" OR treatment OR "adjuvant therapy") AND (use OR uptake OR receipt OR nonuse OR underuse OR initiation OR administration OR utilization OR adherence OR patterns))	2,227,532
#3	TI=("colon cancer" OR "bowel cancer" OR "colonic cancer" OR "colorectal cancer")	95,668
	All combined: #1 AND #2 AND #3	*416*

Supplemental Table S2: Results of the associations of overall comorbidity with adjuvant chemotherapy use

First author	Year	Country	Comorbidity			Effect estimate (95% CI)	Covariates adjusted for								
			Summary	Score			Age	Sex	MS	SES	Edu	Hosp	Year	FS	Others
Charlson comorbidity index (CCI)															
Gupta [29]	2004	USA	CCI	Ref= CCI 0	OR		√	√	√	√					*a,b,j*
			MedPAR	1		**0.59 (0.41-0.87)**									
			(inpatient)	2+		**0.28 (0.19-0.42)**									
			NCH	1		**0.90 (0.72-0.58)**									
			(outpatient)	2+		**0.72 (0.58-0.90)**									
			Outpatient	1		0.95 (0.74-1.22)									
			(all)	2+		0.89 (0.65-1.23)									
McGory [25]	2006	USA	CCI	0	OR	Ref.	√	√		√			√		*a,c*
				1		**0.82 (0.75-0.90)**									
				2		**0.62 (0.54-0.71)**									
				3+		**0.39 (0.32-0.48)**									
Cress [40]	2009	USA	CCI	0	OR	Ref.	√	√	√	√	√				*a,c,j*
				1		0.79 (0.53-1.17)									
				2+		**0.36 (0.19-0.70)**									
Sarfati [48]	2009	NZ	CCI	0	OR	Ref.	√	√							*a*
				1-2		**0.31 (0.14-0.71)**									
				3+		**0.05 (0.01-0.24)**									
Winget [50]	2010	Canada	CCI	0	RR	Ref.	√			√					*j*
				1+		**0.65 (0.55-0.76)**									
Foley [41]	2011	USA	CCI	0	OR	Ref.	√	√		√		√			*a*
				1+		0.89 (0.63-1.28)									
Hu [44]	2011	USA	CCI	0	OR	Ref.	√	√	√	√			√		*a,b,j*
				1		**0.63 (0.57-0.70)**									
				2+		**0.37 (0.33-0.42)**									
Quipourt [46]	2011	France	CCI	Ref=0-1	OR			√							
			<75 years	2+		**0.22 (0.09-0.53)**									
			75+ years	2+		1.15 (0.61-2.15)									
van der Geest [49]	2012	Netherlands	CCI	0	OR	Ref.	√	√		√					*b,d,f,g*
				1		0.83 (0.48-1.41)									
				2+		**0.49 (0.27-0.88)**									
Jorgensen [26]	2014	Australia	CCI	0	OR	Ref.	√	√	√	√			√		*b,d,f*
				1		**0.34 (0.16-0.72)**									
				2		0.75 (0.29-1.94)									
				3+		**0.28 (0.09-0.89)**									
Krarup [27]	2014	Denmark	CCI	0	OR	Ref.	√	√							*d,f,g,h*
				1		**0.86 (0.74-0.99)**									
				2		**0.72 (0.60-0.85)**									
				3+		**0.62 (0.49-0.79)**									

Supplemental Table S2 continues on next page

Supplemental Table S2 (continued): Results of the associations of overall comorbidity with adjuvant chemotherapy use

First author	Year	Country	Comorbidity			Effect estimate (95% CI)	Covariates adjusted for								
			Summary	Score			Age	Sex	MS	SES	Edu	Hosp	Year	FS	Others
Charlson comorbidity index (CCI)															
Becerra [38]	2015	USA	CCI	0	OR	Ref.	√	√		√	√	√	√		*a,c,j*
				1		**0.84 (0.81-0.87)**									
				2+		**0.54 (0.51-0.57)**									
Murphy [45]	2015	USA	CCI	0	RR	Ref.	√						√		*a,c*
				1		1.00 (0.96-1.04)									
				2+		**0.56 (0.35-0.87)**									
Booth [39]	2016	Canada	CCI	0	RR	Ref.	√	√		√		√			*b,e,j*
				1		0.93 (0.85-1.02)									
				2+		**0.67 (0.58-0.78)**									
Hsieh [43]	2016	USA	CCI	nonCCI com.	OR	Ref.	√	√		√			√		*a,b,c,j*
				0		0.86 (0.68-1.08)									
				1		0.96 (0.77-1.19)									
				2+		**0.62 (0.42-0.91)**									
Guerrero [42]	2017	USA	CCI	0	OR	Ref.	√								*a,c,g*
				1-2		0.94 (0.57-1.57)									
				3-4		0.51 (0.22-1.20)									
				5+		0.25 (0.06-1.03)									
van den Broek [37]	2017	Netherlands	CCI	0	OR	Ref.									
				1		**0.31 (0.16-0.64)**									
				2+		**0.08 (0.04-0.14)**									
Yamano [10]	2017	Japan	CCI	Ref=0	OR										
			<75 years	1		**0.59 (0.39-0.88)**									
				2+		**0.36 (0.21-0.63)**									
			75+ years	1		0.79 (0.41-1.53)									
				2+		0.92 (0.42-1.99)									
Boakye [28]	2019	Germany	CCI	0	OR	Ref.	√	√	√		√	√	√	√	*b,d*
				1		**0.43 (0.22–0.84)**									
				2		**0.34 (0.16–0.72)**									
				3+		**0.30 (0.14–0.64)**									
Sarasqueta [47]	2019	Spain	CCI	0	OR	Ref.	√	√	√	√	√	√		√	*b,d*
				1		0.80 (0.50-1.40)									
				2+		0.60 (0.30-1.20)									
Boyle [9]	2020	UK	RCS CCI	0	OR	Ref.	√	√			√			√	*b,d,f,j*
				1		**0.80 (0.72-0.90)**									
				2+		**0.50 (0.44-0.58)**									
Minicozzi [7]	2020	European Countries [a]	CCI	0-1	OR	Ref.	√	√							*d*
				2+		**0.51 (0.39-0.65)**									

Supplemental Table S2 continues on next page

Supplemental Table S2 (continued): Results of the associations of overall comorbidity with adjuvant chemotherapy use

First author	Year	Country	Comorbidity			Effect estimate (95% CI)	Covariates adjusted for								
			Summary	Score			Age	Sex	MS	SES	Edu	Hosp	Year	FS	Others
Adult comorbidity evaluation (ACE-27)															
Kahn [30]	2010	USA	ACE-27	Ref=None	OR										
			<u>≤75 years</u>	Mild		0.63 (0.31-1.32)									
				Moderate		0.53 (0.22-1.26)									
				Severe		**0.27 (0.12-0.63)**									
			<u>75+ years</u>	Mild		1.00 (0.44-2.29)									
				Moderate		0.77 (0.31-1.93)									
				Severe		0.66 (0.25-1.75)									
Hoeben [36]	2013	Netherlands	ACE-27	None	OR										
				Mild		0.79 (0.50-1.25)									
				Moderate		0.78 (0.49-1.26)									
				Severe		**0.42 (0.24-0.73)**									
Fleming [51]	2014	USA	ACE-27	None	OR	Ref.	√								
				Mild		1.44 (0.81-2.56)									
				Moderate		1.09 (0.54-2.21)									
				Severe		0.90 (0.47-1.74)									
Comorbidity count															
Lemmens [52]	2005	Netherlands	Count	0	OR	Ref.	√	√							
				1		0.80 (0.60-1.10)									
				2+		**0.50 (0.30-0.80)**									
van Steenbergen [51]	2010	Netherlands	Count	0	OR	Ref.	√	√		√		√	√		*b*
				1		**0.70 (0.50-0.90)**									
				2+		**0.40 (0.30-0.60)**									
van den Broek [37]	2017	Netherlands	Count	0	OR										
				1		**0.26 (0.15-0.45)**									
				2+		**0.07 (0.04-0.14)**									

Abbreviations: *ACE-27*=Adult comorbidity evaluation-27; *CCI*=Charlson comorbidity index; *CI*=Confidence interval; *Com*=Comorbidity; *Educ*=Education level; *FS*=Functional or performance status; *Hosp*=Hospital characteristics; *MedPAR*=Medicare Provider Analysis and Review; *NCH*=National Claim History; *OR*=Odds ratio; *RCS*=Royal College of Surgeons; *Ref*=Reference group; *RR*=Relative risk; *SES*=Socioeconomic status; *UK*=United Kingdom; *USA*=United States of America; *Year*=Year of diagnosis

[a] Eight European cancer registries (Belgium, Estonia, Germany, Italy, Poland, Portugal, Slovenia, and Spain)

Others (*a*=race or ethnicity; *b*=tumor grade, sub-site, or size; *c*=health insurance type; *d*=type of surgical technique; *e*=length of hospital stay after colon cancer surgery; *f*=urgency of surgery; *g*=anastomotic leakage or postoperative complications; *h*=blood transfusion; *j*=state or region of residence)

Statistically significant results are highlighted in bold (OR/RR < 1.00 indicates lower likelihood of receiving chemotherapy)

Supplemental Table S3: Results of the associations of individual comorbid conditions with adjuvant chemotherapy use

First author	Year	Country	Comorbidities (yes vs no) [c]	OR (95% CI)	Covariates adjusted for [d]								
					Age	Sex	MS	SES	Edu	Hosp	Year	FS	Others
Gupta [29]	2004	USA	Dementia	**0.22 (0.13-0.36)**	√	√	√	√					*a,b,d*
Lemmens [52]	2005	Netherlands	COPD	**0.30 (0.10-0.90)**	√	√							
Gross [55]	2007	USA	Diabetes	**0.81 (0.68-0.97)**	√	√	√	√					*a,b,d*
			Heart failure	**0.49 (0.40-0.60)**									
			COPD	**0.83 (0.70-0.99)**									
Baillargeon [57]	2011	USA	Dementia	**0.31 (0.26-0.38)**	√	√	√	√			√		*a,d*
Carpenter [54]	2012	USA	Diabetes	0.86 (0.71-1.05)	√	√			√		√		*a,b*
			Renal disease	0.91 (0.44-1.86)									
			Dementia	0.66 (0.38-1.17)									
Rodriguez [35] [a,b]	2013	USA	Diabetes	1.12 (0.50-2.53)									
			Heart failure	**0.16 (0.04-0.63)**									
			COPD	0.71 (0.27-1.89)									
Zanders [56]	2013	Netherlands	Diabetes	**0.70 (0.50-0.90)**	√	√					√		
			Stroke	**0.60 (0.40-0.90)**									
			COPD	**0.60 (0.50-0.80)**									
Fleming [51]	2014	USA	Myocardial infarction	0.66 (0.32-1.34)	√								
			Heart failure	0.70 (0.38-1.27)									
			COPD	0.71 (0.45-1.12)									
			Dementia	**0.11 (0.01-0.90)**									
Chen [14] [b]	2017	USA	Dementia	**0.49 (0.39-0.60)**									
Boakye [28] [a]	2019	Germany	Diabetes	**0.41 (0.22-0.78)**	√	√	√		√	√	√	√	*b,c*
			Myocardial infarction	0.52 (0.21-1.24)									
			Heart failure	**0.21 (0.06-0.68)**									
			Stroke	**0.38 (0.15-0.97)**									
			Renal disease	**0.18 (0.07-0.48)**									
			COPD	**0.28 (0.12-0.65)**									

Abbreviations: *CI*=Confidence interval; *COPD*=Chronic obstructive pulmonary disease; *Educ*=Education level; *FS*=Functional or performance status; *Hosp*=Hospital characteristics; *OR*=Odds ratio; *SES*=Socioeconomic status; *USA*=United States of America; *Year*=Year of diagnosis

[a] Sample included stage II high-risk patients.

[b] ORs are crude estimates

[c] Reference groups were absence of respective comorbidities except for the study by Boakye et al (2019), wherein the reference group was patients with no Charlson comorbidity.

[d] Adjusted ORs were also adjusted for all other comorbid conditions

Others (*a*=race or ethnicity; *b*=tumor grade, sub-site, or size; *c*=type of surgical technique; *d*=state or region of residence)

Statistically significant results are highlighted in bold (OR/RR < 1.00 indicates lower likelihood of receiving chemotherapy)

Supplemental Table S4: Quality assessment of the included studies using the Newcastle-Ottawa scale

Study	Representativeness of study sample		Comorbidity assessment				Chemotherapy assessment		Control of confounding	Total
Author (year)	Selection	Age, sex, or comorbidity	≤1 month of CC diagnosis	Physician-diagnosed	Medical records	Missing	Surgical resection	Missing	At least age and year of diagnosis	
Gupta (2004) [29]	1	1	1	1	0	0	1	0	0	5
Lemmens (2005) [52]	1	1	1	1	1	1	1	1	0	8
McGory (2006) [25]	1	1	1	1	0	0	1	1	1	7
Gross (2007) [55]	1	1	1	1	0	1	1	1	1	8
Cress 2009 [40]	1	1	0	1	1	1	1	1	1	8
Sarfati (2009) [48]	0	1	1	1	1	1	1	1	0	7
Kahn (2010) [30]	0	1	1	1	1	1	1	1	0	7
van Steenbergen (2010) [51]	1	1	1	1	0	0	0	1	1	6
Winget (2010) [50]	1	1	1	1	0	1	1	1	0	7
Baillargeon (2011) [57]	1	1	1	1	0	1	0	1	1	7
Foley (2011) [41]	1	1	1	1	0	1	1 [a]	1	0	7
Hu (2011) [44]	1	1	1	1	0	1	1	1	1	8
Quipourt (2011) [46]	1	1	1	1	0	1	1	0	0	6
Carpenter (2012) [54]	1	1	0	1	0	1	1	1	1	7
van der Geest (2012) [49]	1	1	0	1	0	1	1	1	0	6
Hoeben (2013) [36]	0	1	1	1	1	0	1	1	0	6
Rodriguez (2013) [35]	0	0	0	1	0	1	1	1	0	4
Zanders (2013) [56]	1	1	1	1	0	1	0	1	1	7
Fleming (2014) [51]	1	0	0	1	0	1	0	1	0	4
Jorgensen (2014) [26]	1	0	0	1	0	1	1	1	1	6
Krarup (2014) [27]	1	1	1	1	0	1	1	1	0	7
Becerra (2015) [38]	1	1	0	1	0	1	1	1	1	7
Murphy (2015) [45]	1	1	1	1	1	1	1	1	1	9
Booth (2016) [39]	0	1	1	1	0	1	1	1	0	6
Hsieh (2016) [43]	1	1	0	1	0	1	1	0	1	6
Chen (2017) [14]	1	1	1	1	0	0	1	0	0	5
Guerrero (2017) [42]	1	1	0	1	0	1	1	1	0	6
van den Broek (2017) [37]	1	1	1	1	1	1	1	1	0	8
Yamano (2017) [10]	1	1	0	1	0	1	1	1	0	6
Boakye (2019) [28]	0	1	1	1	1	1	1	1	1	8
Sarasqueta (2019) [47]	1	1	0	1	0	1	1	1	0	6
Boyle (2020) [9]	1	1	1	1	0	1	1	1	0	7
Minicozzi (2020) [7]	1	1	1	1	0	0	1	1	0	6
Scoring	**1**	**1**	**1**	**1**	**1**	**<10%, 1**	**1**	**<10%, 1**	**1**	**9**

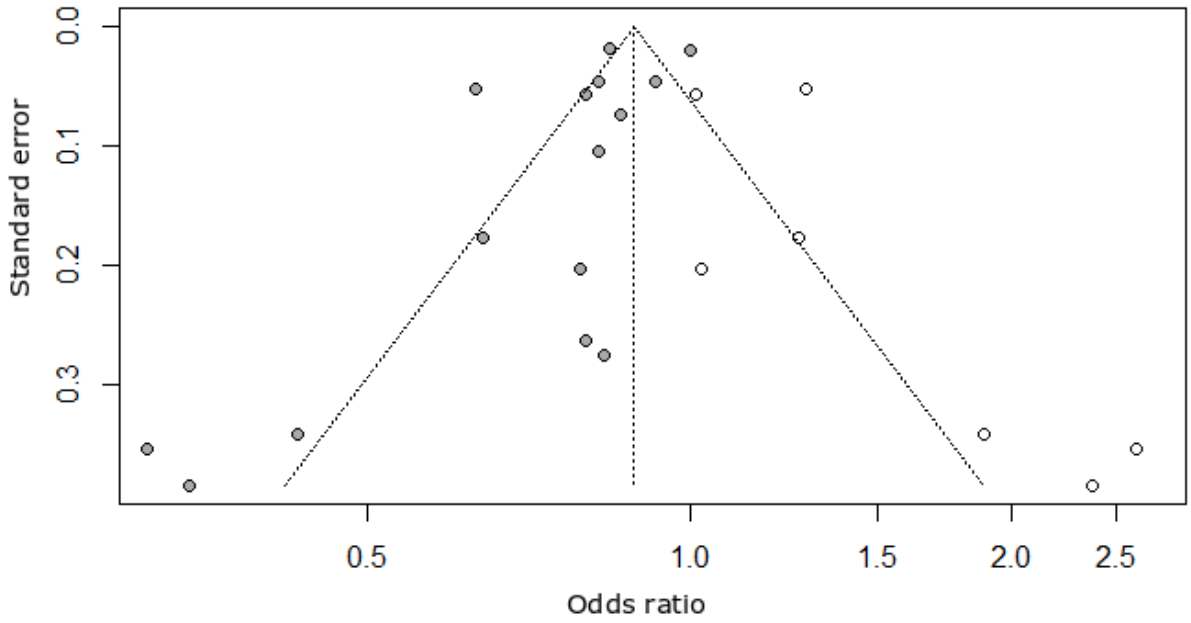
Abbreviation: *CC*=Colon cancer

[a] >98% had surgical resection of colon cancer

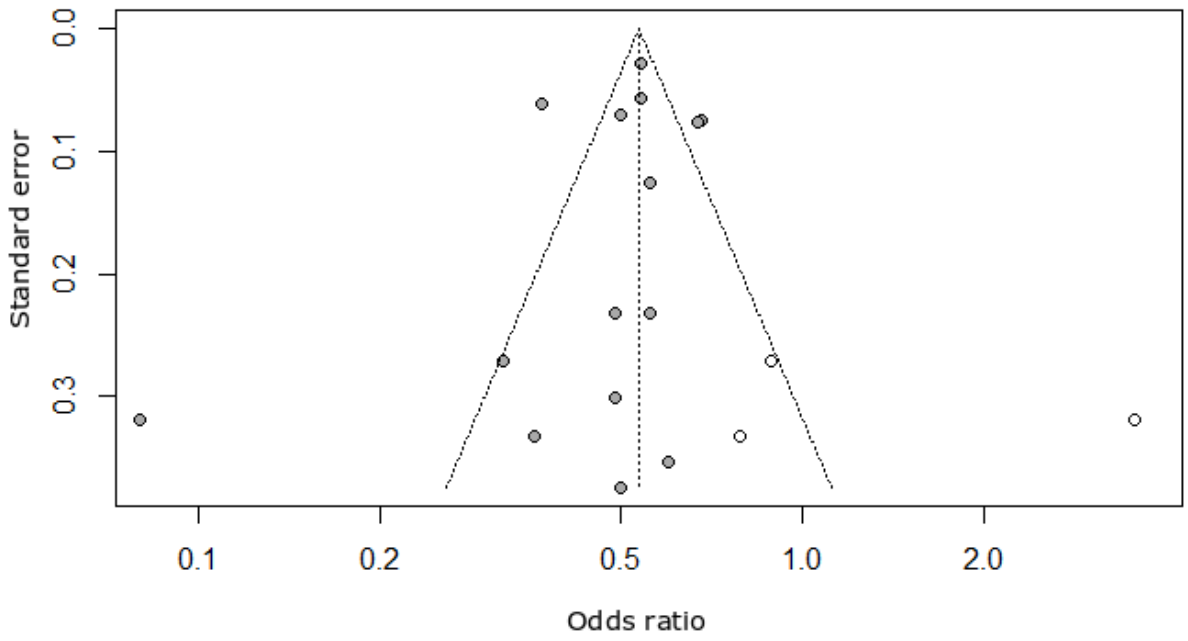
For studies recruiting participants within two calendar years of diagnosis, it was assumed that chemotherapy use was comparable by year of diagnosis.

Supplemental Figure S1: Funnel plot assessing potential publication bias in studies on Charlson comorbidity index score and chemotherapy use

A: CCI score 1 versus 0



B: CCI score 2+ versus 0



Abbreviation: *CCI*=Charlson comorbidity index

Note: Empty circles indicate unpublished potentially eligible studies artificially created by the Trim-and-Fill method (OR < 1.00 means lower likelihood of receiving chemotherapy)

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5, Suppl. Tab 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5-7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6-7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6-7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8, Fig 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8, Tab 1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Tab 1, Suppl. Tab 4
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9-11, Suppl. 2-4
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10-11, Fig 2-4
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	10-12, Suppl. Fig 1
Additional analysis	23	Give results of additional analyses, if done (e.g., subgroup analyses, meta-regression [see Item 16]).	11-12, Tab 3
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	13-16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	18
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	19

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097